Learning by Shifting:
Temporal View Construction for Time Series Contrastive Learning

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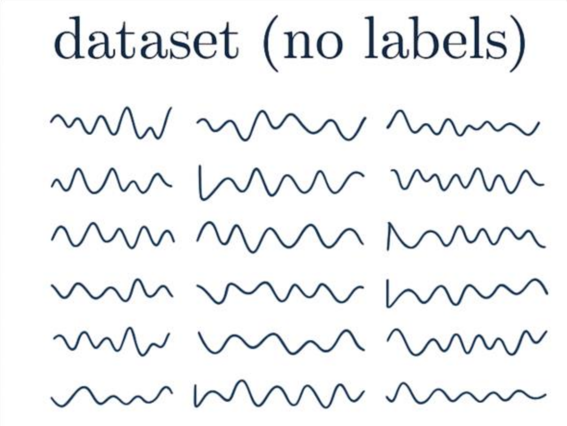
Time Series

Self-Supervised Learning Motivation



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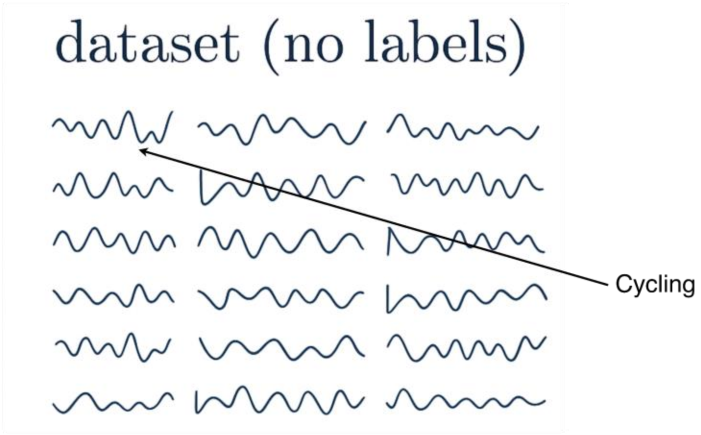

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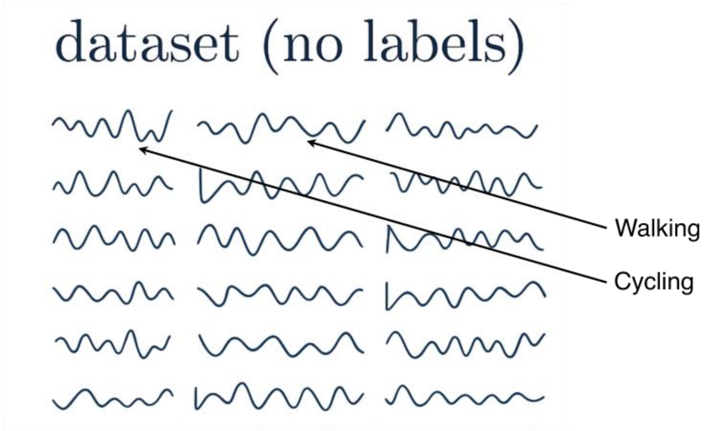

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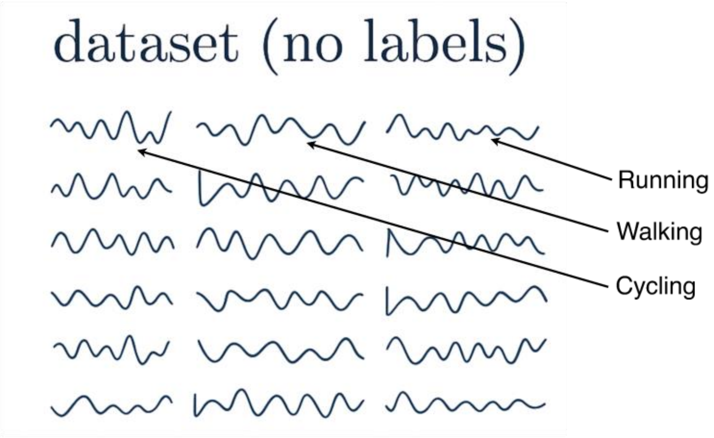

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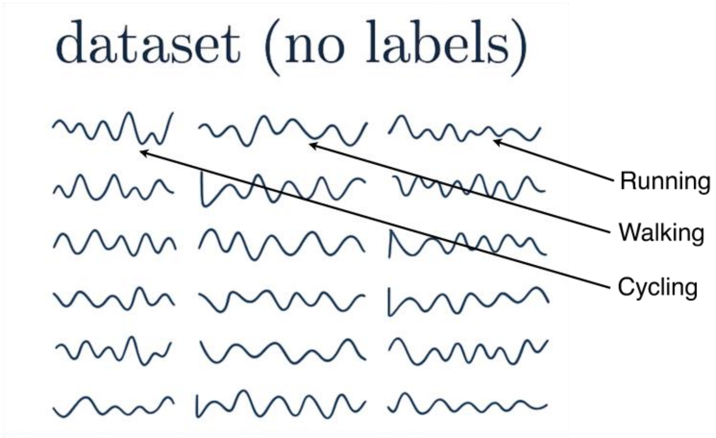
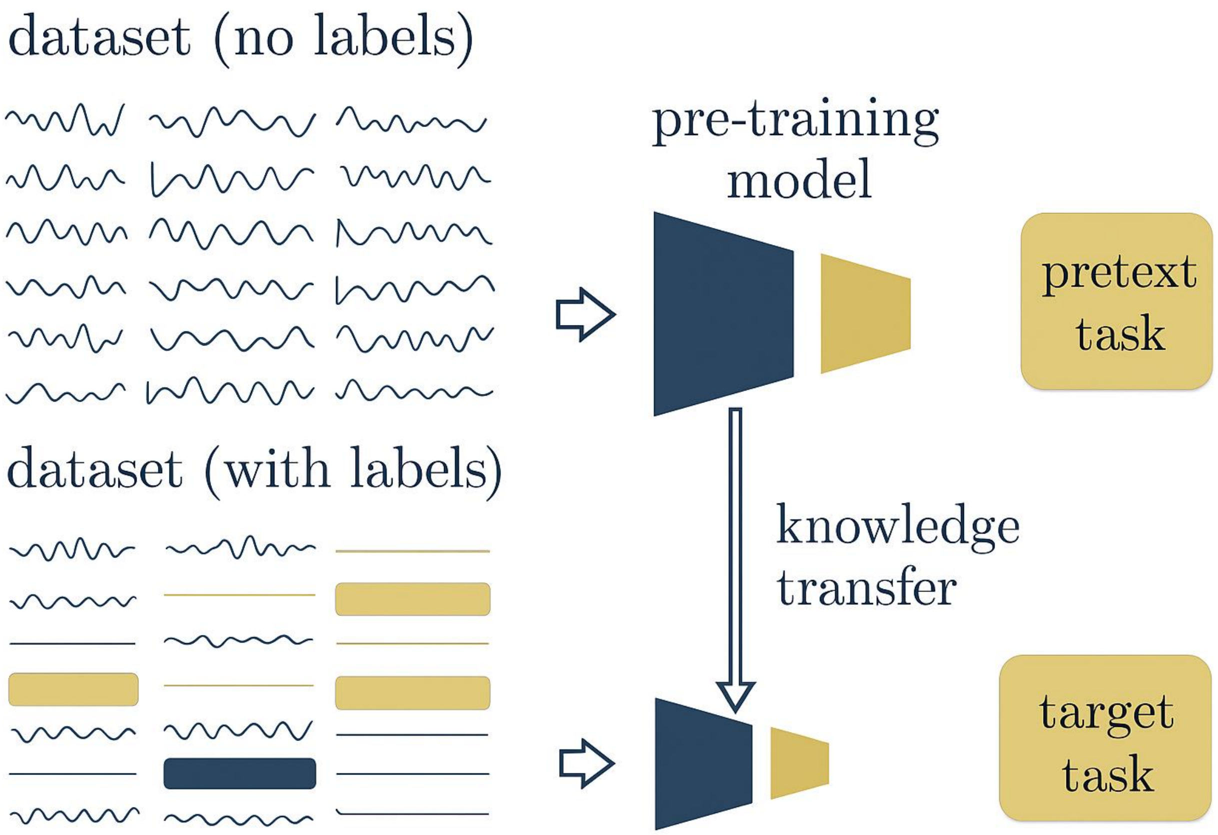


Supervised learning
- Relies on large labeled datasets
- Expensive and domain-specific annotations
- Limited generalization

Self-Supervised Learning Motivation



Supervised learning

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- Limited generalization

Self-Supervised learning

- Learns representations directly from data
- Scalable (unlabeled data is abundant)
- Improves transferability to downstream tasks

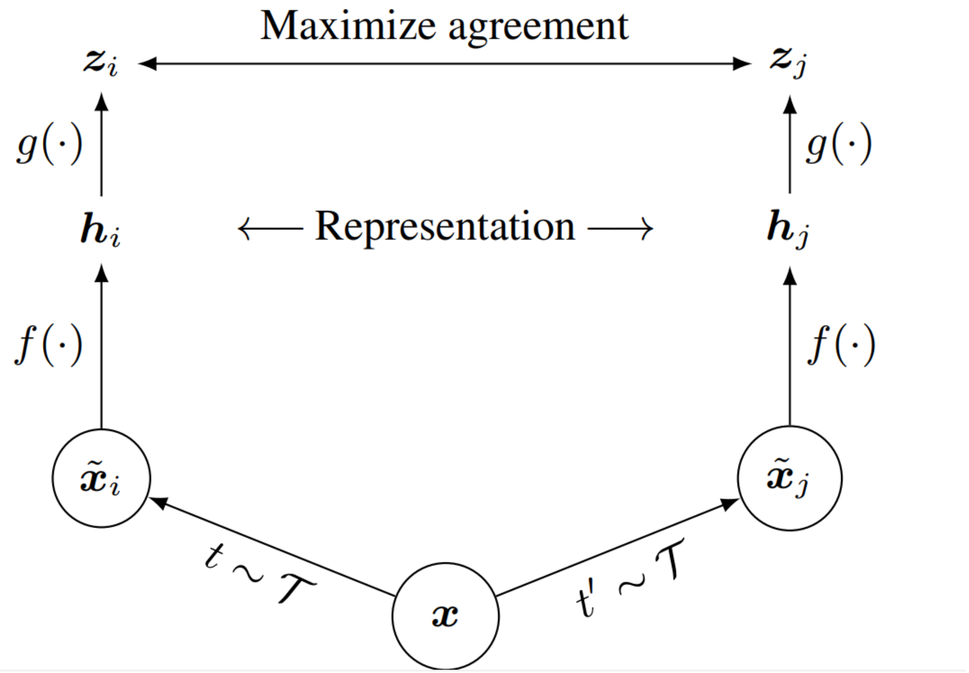
Self-Supervised Learning – Pretext Task

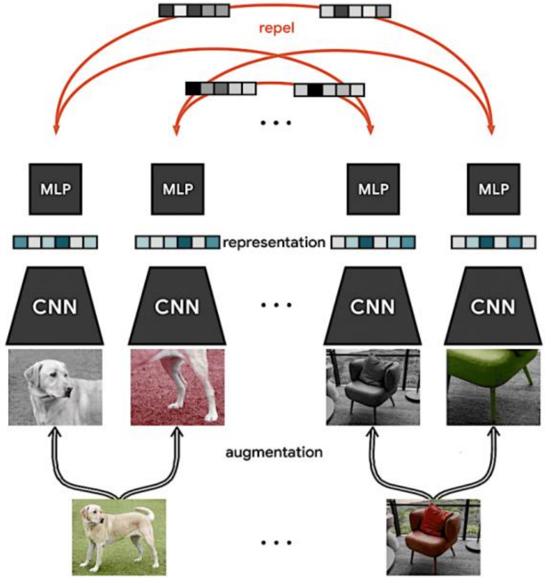
source: image generated by Microsoft copilot

Pretext Tasks

- Jigsaw puzzle
- Inpainting
- Colorization
- Rotation
- Reconstruction

Contrastive Representation Learning

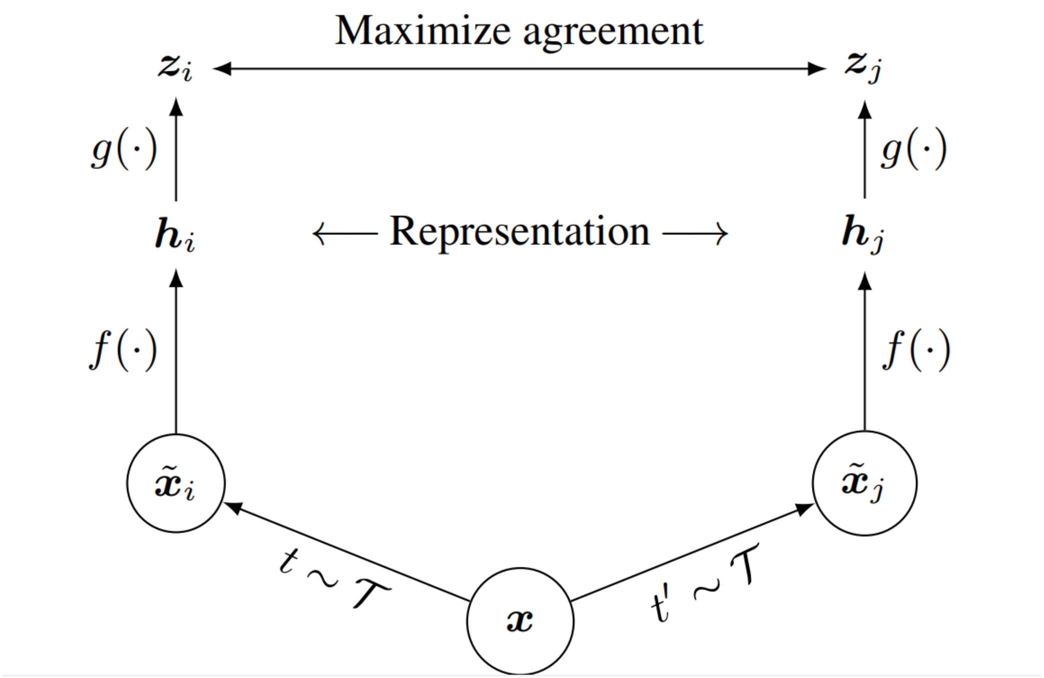


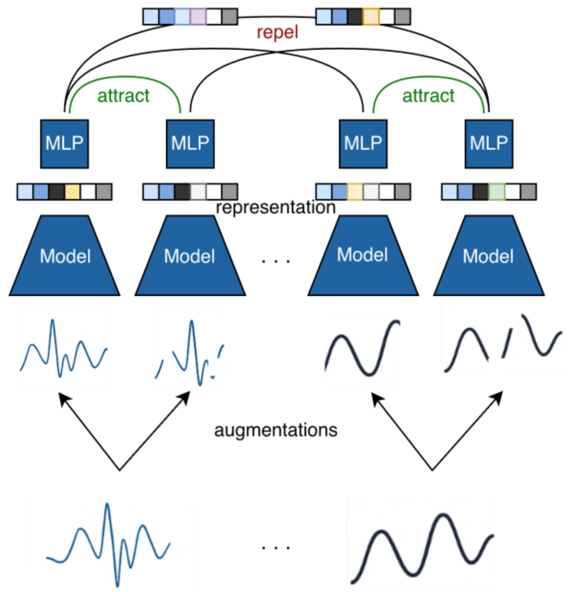


SimCLR, A Simple Framework for Contrastive Learning of Visual Representations, ICML 2020

Contrastive Representation Learning

Time Series





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Contrastive Learning in Time Series

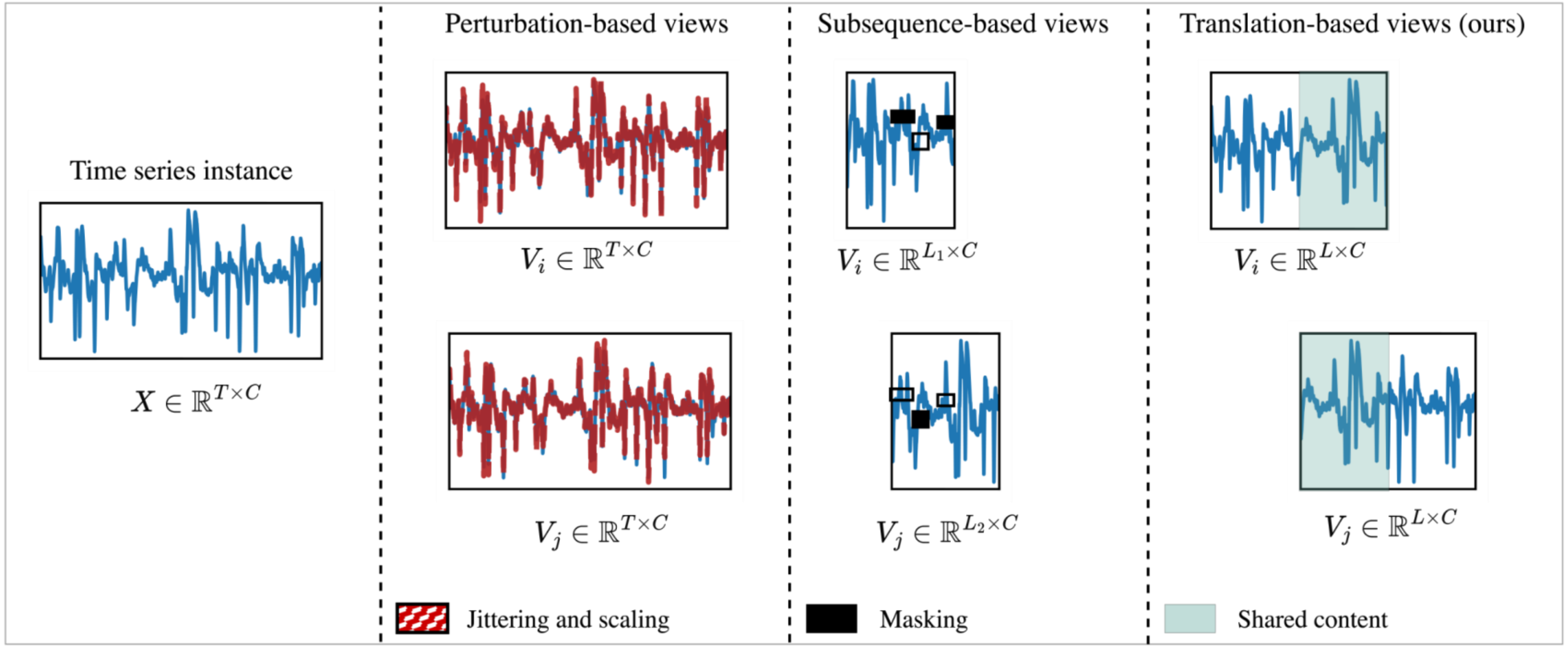
- Hand crafted augmentations and masking
- Positives selection using dynamic time warping

Consequence
- Domain-specific inductive biases
- Additional computational overhead

Success criteria
- No augmentation and domain specific assumptions
- Improved accuracy and generalization
- Reduction in training time
- Compact (clusters) and transferable representations

Looking at the temporal information

View construction for in time series

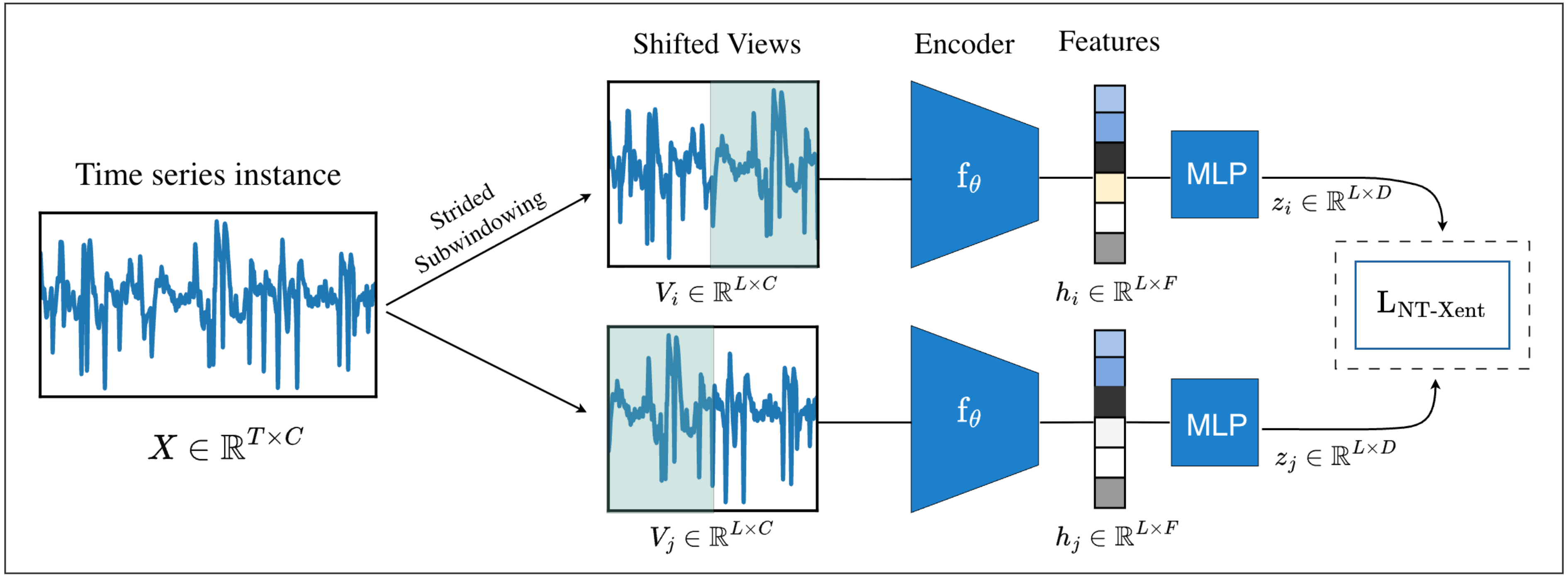


Temporal Shift

- Preserves label maximally
- Retain discriminative structure naturally
- Minimal domain assumptions

Learning by Shifting

Temporal View Construction for Time Series Contrastive Learning



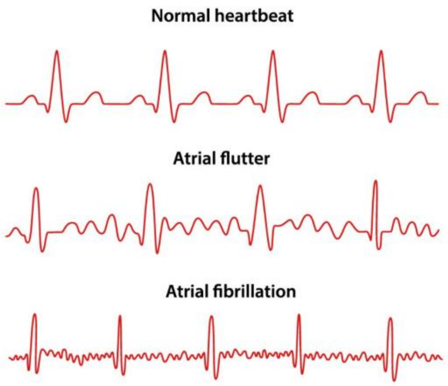
- More efficient training: With the default 50% overlap, each view contains only 2/3 of the original sequence, reducing computation and contributing to faster training

Dataset

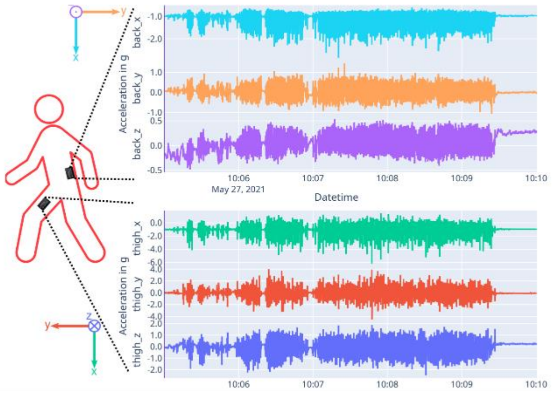
Six (6) Large Datasets with >20*k* instance each

Dataset	Train set	Test set	Subject	Sequence Length	Channel	Class
ECG	59,922	16,645	23	2,500	2	4
HARTH	23,025	2,785	22	500	6	18
PAMAP2	51,192	11,590	9	100	52	18
SKODA	22,587	5,646	1	50	64	11
SLEEP	25,612	8,910	20	3,000	1	5
WISDM2	134,614	14,421	29	40	3	6

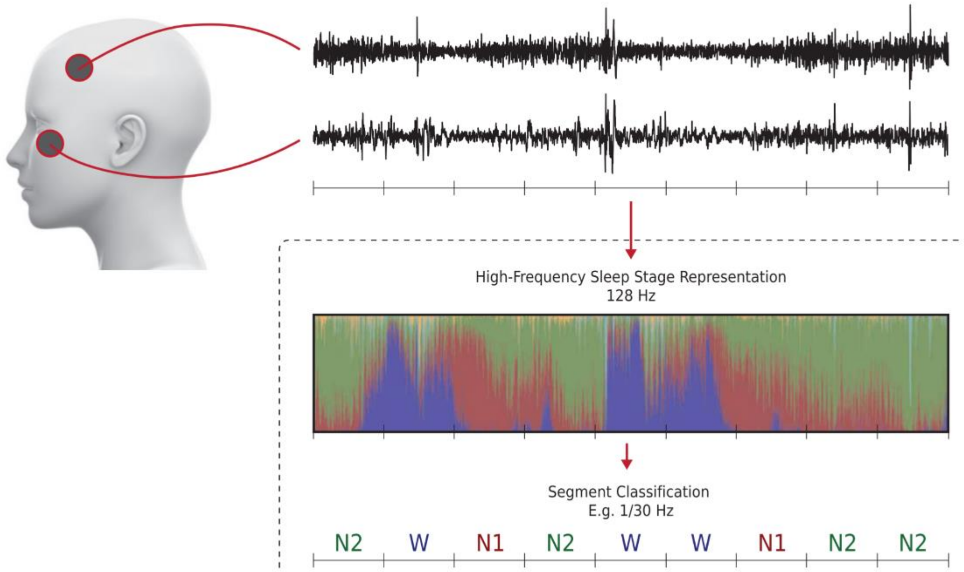
ECG



HARTH



SleepEEG



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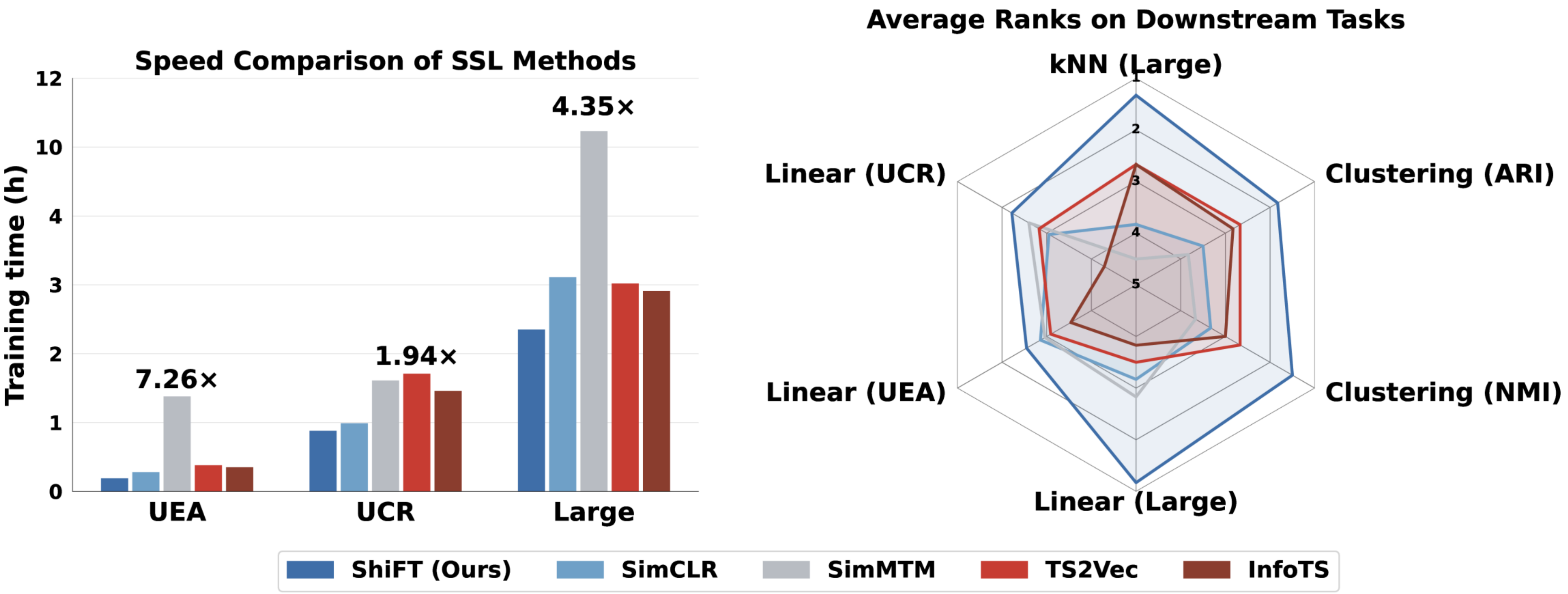
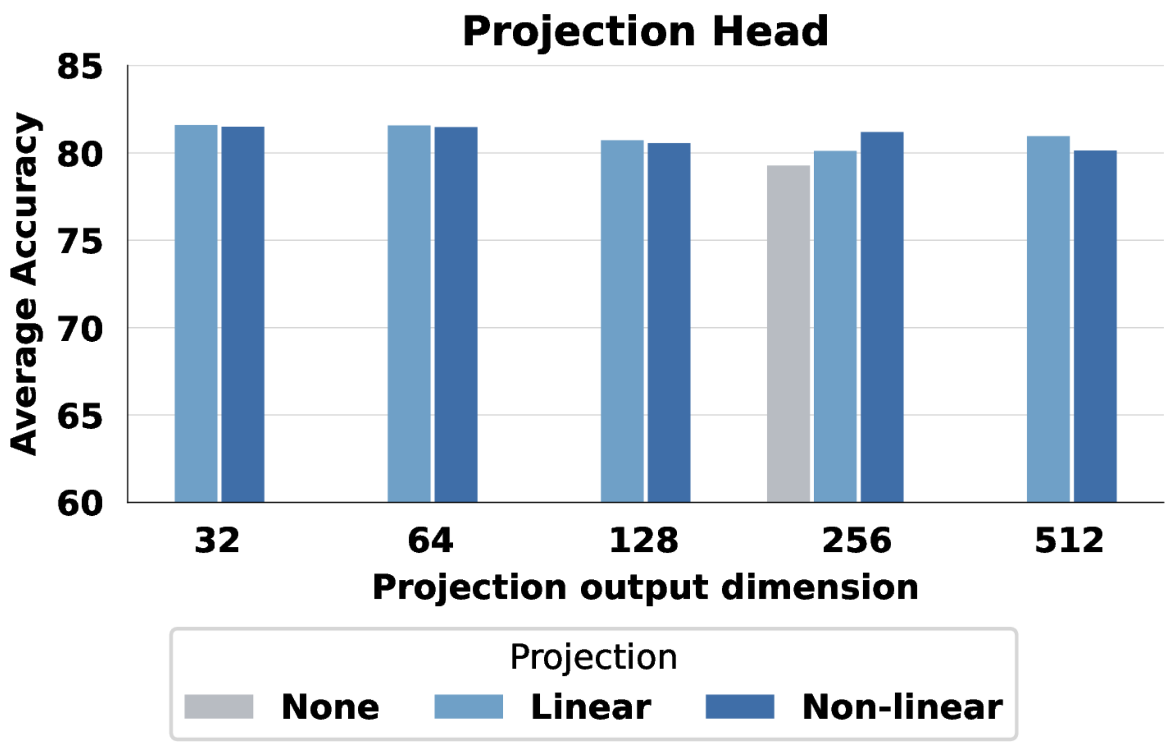
Training efficiency and downstream performance



Learning by Shifting

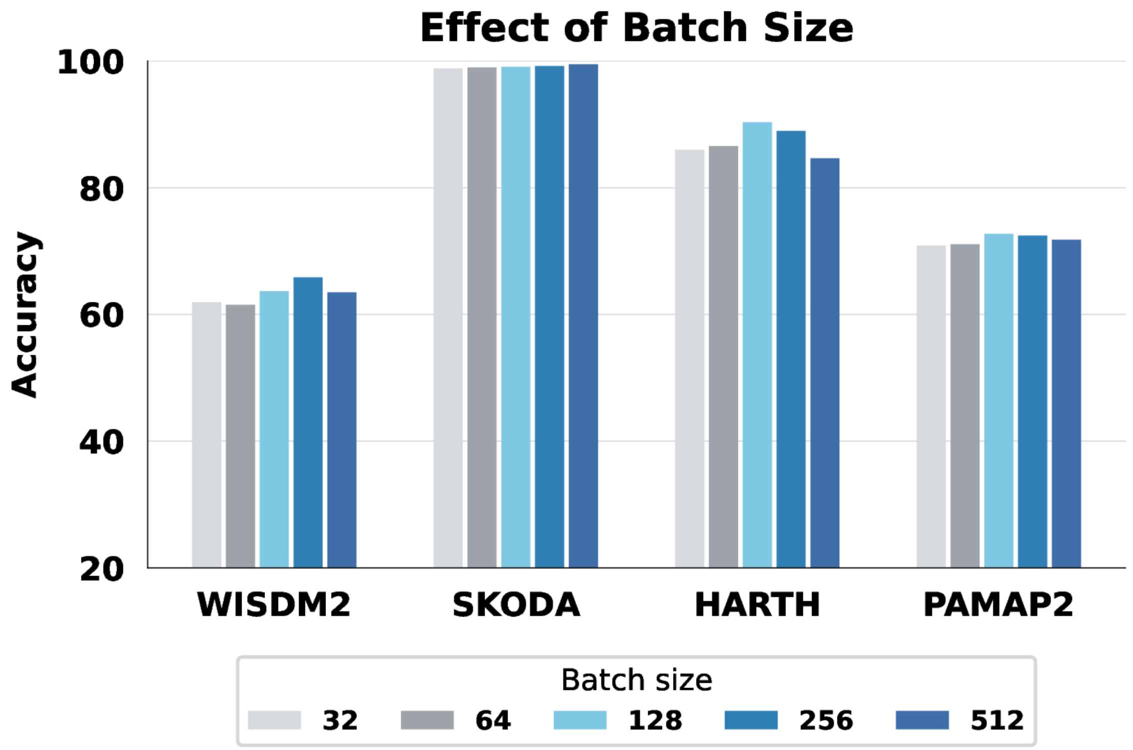
Projection Head Design



- Projection heads help
- Head complexity matters little
- Main bottleneck is view construction

Learning by Shifting

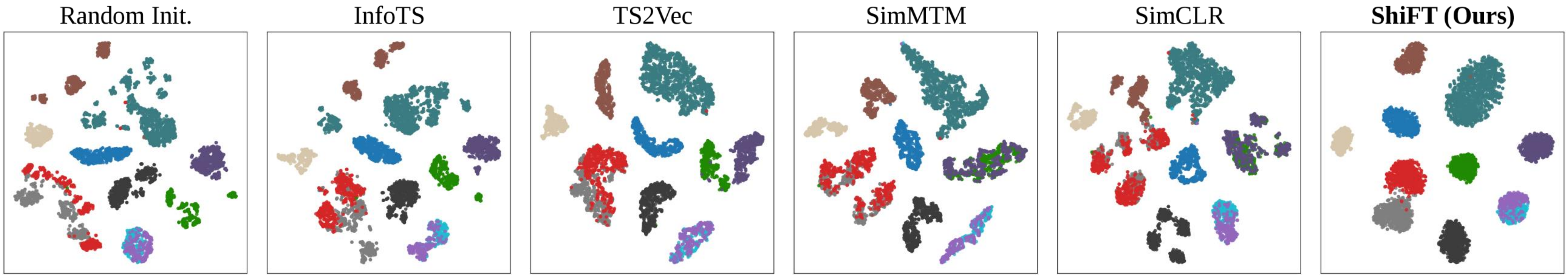
Batch Size in Time Series Contrastive Learning



- Larger batches do not consistently help
- Moderate batch sizes work best
- False negatives limit large batches

Learning by Shifting

t-SNE Visualisation of Learned Embeddings



- Well-separated and compact clusters across all activity classes
- Cleanly separates the challenging red and gray classes
- Shift invariance is *almost* all you need!

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Read more!

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